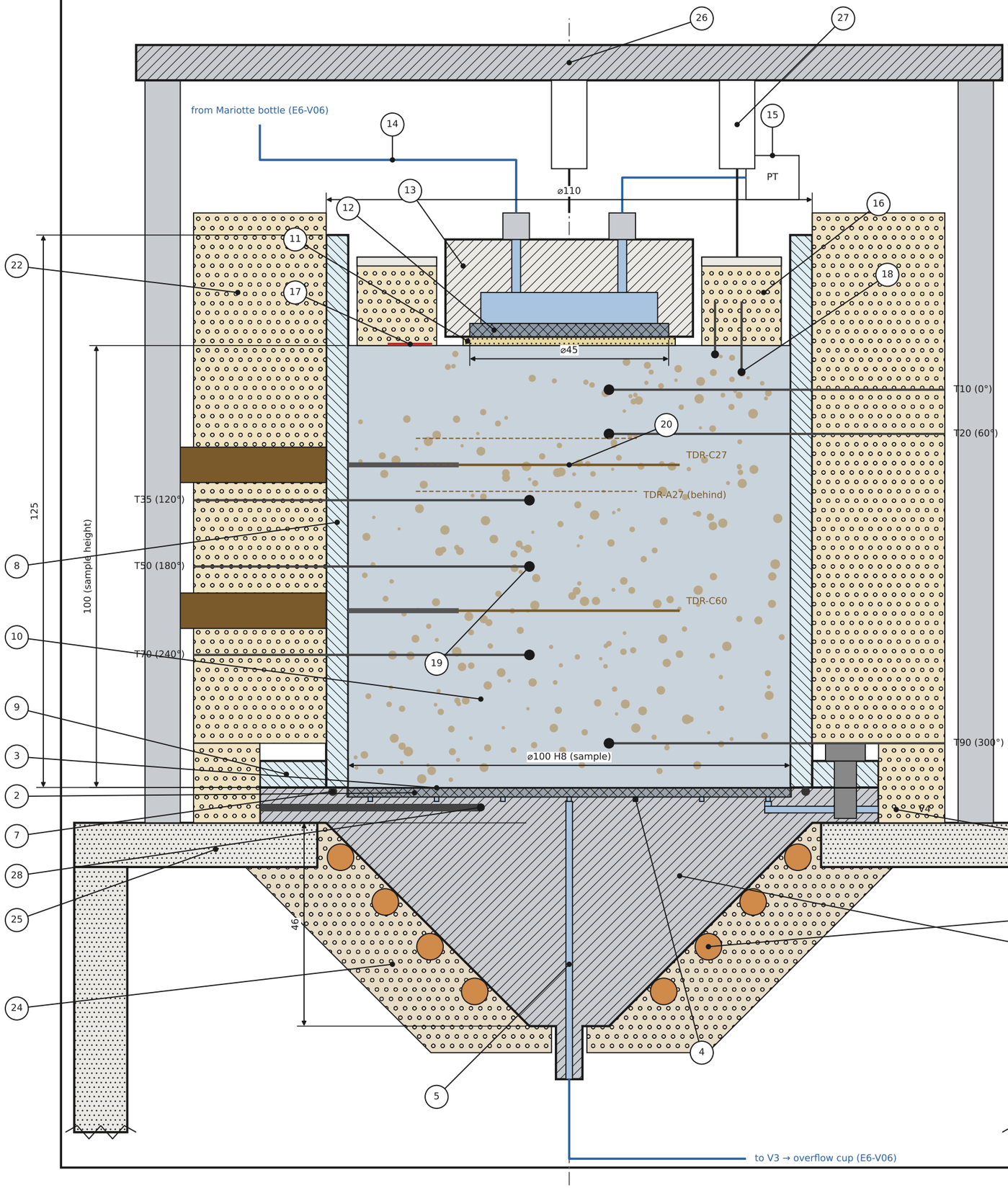


Parts list

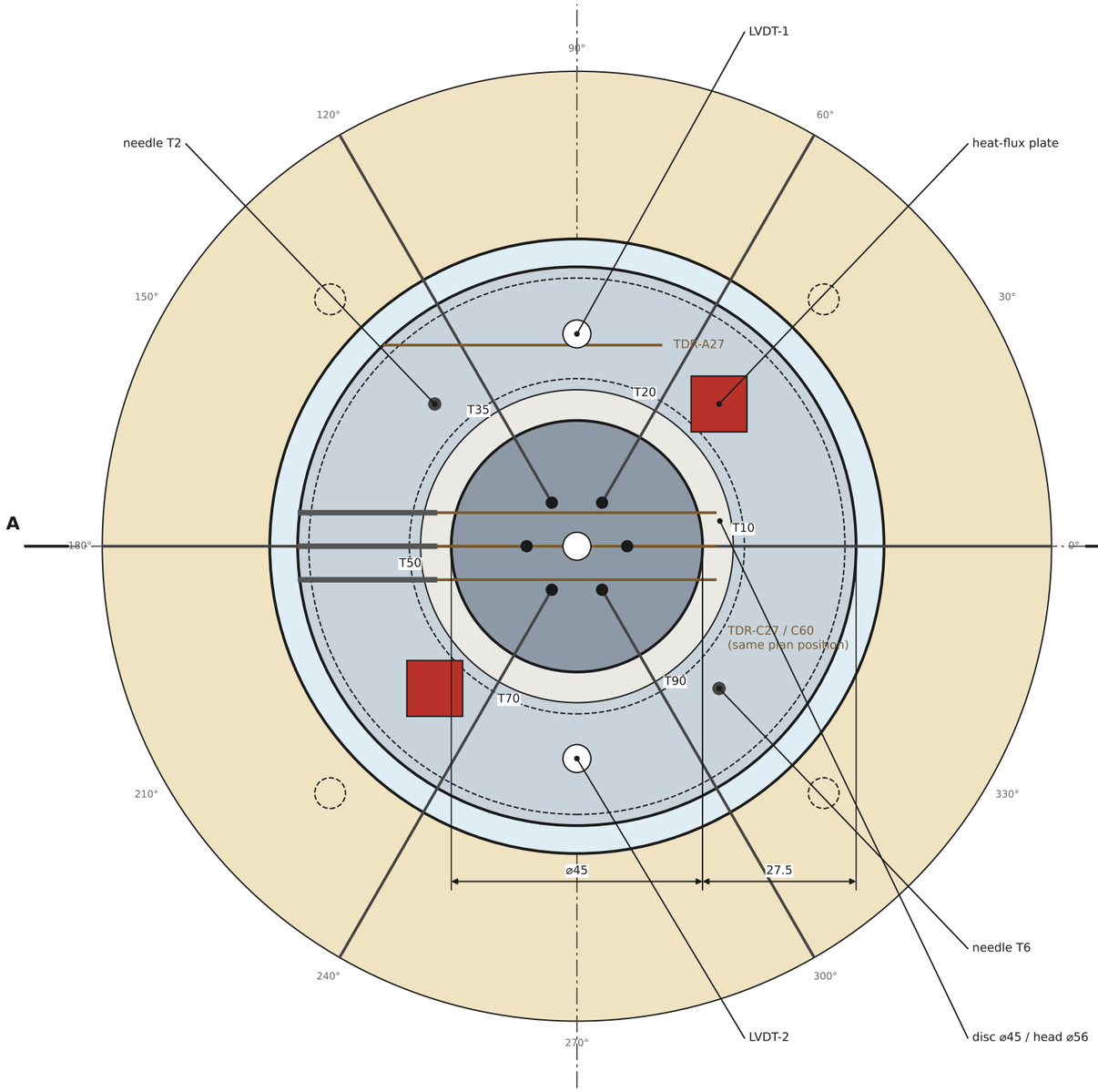
1. cone outlet body, Al 6082, solid, 45° (E6-V04)
2. perforated support plate AISI 316, t 2, holes ø1.5 / pitch 4
3. nylon membrane 20 µm, ø104, clamped by tube end face
4. drainage grooves 1×1: 12 radial + 3 rings (r 15/30/45)
5. axial bore ø1.5 → nozzle ø6 → PTFE 1/16" → valve V3
6. air-purge port from ring groove r 45, valve V4
7. O-ring ø107×2 NBR, face seal
8. PMMA tube ø110/100×125, inner wall sand-coated (epoxy)
9. PMMA flange ø140×6, bonded; 4× M5 into cone rim
10. soil sample ø100×100, packed in 10 mm lifts
11. contact sand 0.1–0.3 mm, t 2, ø48, placed dry
12. sintered disc ø45×3, stainless (as Mini Disk / AMDI)
13. AMDI head body, PETG – low-profile variant, 2 ports
14. supply line from Mariotte bottle on balance, FEP 4/2
15. diff. pressure sensor at disc ±2.5 kPa (+ air purge)
16. floating insulation ring XPS ø96/60×20, PETG frame
17. heat-flux sensor 10×10 mm, 2× at r 36 (45°, 225°)
18. thermistor needles in ring, 2 and 6 mm (135°, 315°)
19. wall thermistor ø1.6, tip at r 9, 6× helix at 60°
20. centre TDR C27, C60: 3 rods ø1.6×75, 25 mm insulated
21. annulus TDR A27: chord y 36, 3 rods ø1.6×50, stacked
22. side insulation Kaiflex t 30
23. Cu coil ø6, 4 turns on cone → circulating bath
24. cone insulation, PU foam shell t 20
25. stand plate POM t 10, 3 legs
26. LVDT bridge, Al 20×8 on posts from stand plate
27. LVDT ±10 mm: 0 – head, 1/2 – ring (90°, 270°)
28. thermistor in cone, 4.5 mm below membrane

z = 0: top face of membrane = bottom of sample.
Sensor depths refer to the initial surface z = 100.
Item 21 lies behind the section plane (dashed).

axisymmetric section; off-plane items revolved into the section plane (angles: E6-V02)

Column assembly – section A–A

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	**1 : 1**	drawing no.	**E6-V01**	sheet 1 / 7



Sensor schedule

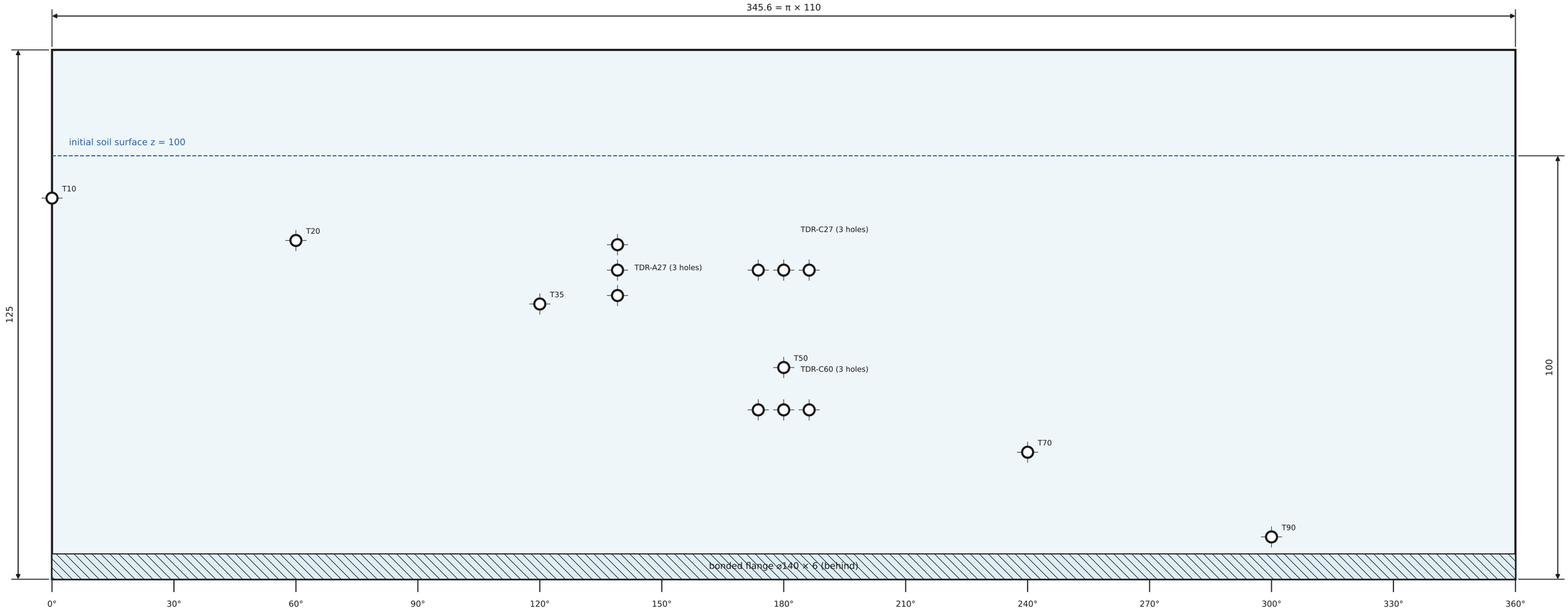
ID	type	angle °	r / y	depth	z
T10	thermistor, wall	0	9	10	90
T20	thermistor, wall	60	9	20	80
T35	thermistor, wall	120	9	35	65
T50	thermistor, wall	180	9	50	50
T70	thermistor, wall	240	9	70	30
T90	thermistor, wall	300	9	90	10
T2r	thermistor needle, ring	135	36	2	floats
T6r	thermistor needle, ring	315	36	6	floats
Tdisc	thermistor, disc/sand	180	7	0	floats
Tcone	thermistor, cone	180	20	–	-4.5
TDR-C27	TDR 3×ø1.6×75, 25 ins.	180	y=0	27	73
TDR-A27	TDR 3×ø1.6×50	chord	y=36	27	73
TDR-C60	TDR 3×ø1.6×75, 25 ins.	180	y=0	60	40
HF1	heat-flux 10×10	45	36	0	floats
HF2	heat-flux 10×10	225	36	0	floats
LVDT-0	LVDT ±10 mm	–	0	–	top
LVDT-1	LVDT ±10 mm	90	38	–	top
LVDT-2	LVDT ±10 mm	270	38	–	top

depth = mm below the initial surface; z = height above the membrane.
Ring sensors float with the surface: they record heave / thaw settlement via LVDT-1/2 and keep a fixed depth below the moving surface.

angle 0° = +x (right half of section A–A), counter-clockwise seen from above

Sensor layout – plan view

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	**1 : 1**	drawing no.	**E6-V02**	sheet 2 / 7



Hole table

ID	angle °	arc mm	z	hole	direction	purpose
T10	0.0	0.0	90	⌀2.0	radial	thermistor ⌀1.6 + gland
T20	60.0	57.6	80	⌀2.0	radial	thermistor ⌀1.6 + gland
T35	120.0	115.2	65	⌀2.0	radial	thermistor ⌀1.6 + gland
T50	180.0	172.8	50	⌀2.0	radial	thermistor ⌀1.6 + gland
T70	240.0	230.4	30	⌀2.0	radial	thermistor ⌀1.6 + gland
T90	300.0	288.0	10	⌀2.0	radial	thermistor ⌀1.6 + gland
TDR-C27.1	186.3	178.8	73	⌀2.0	parallel to x	TDR rod ⌀1.6
TDR-C27.2	180.0	172.8	73	⌀2.0	parallel to x	TDR rod ⌀1.6
TDR-C27.3	173.7	166.8	73	⌀2.0	parallel to x	TDR rod ⌀1.6
TDR-A27.1	139.1	133.5	67	⌀2.0	parallel to x (41° to wall)	TDR rod ⌀1.6
TDR-A27.2	139.1	133.5	73	⌀2.0	parallel to x (41° to wall)	TDR rod ⌀1.6
TDR-A27.3	139.1	133.5	79	⌀2.0	parallel to x (41° to wall)	TDR rod ⌀1.6
TDR-C60.1	186.3	178.8	40	⌀2.0	parallel to x	TDR rod ⌀1.6
TDR-C60.2	180.0	172.8	40	⌀2.0	parallel to x	TDR rod ⌀1.6
TDR-C60.3	173.7	166.8	40	⌀2.0	parallel to x	TDR rod ⌀1.6

Notes

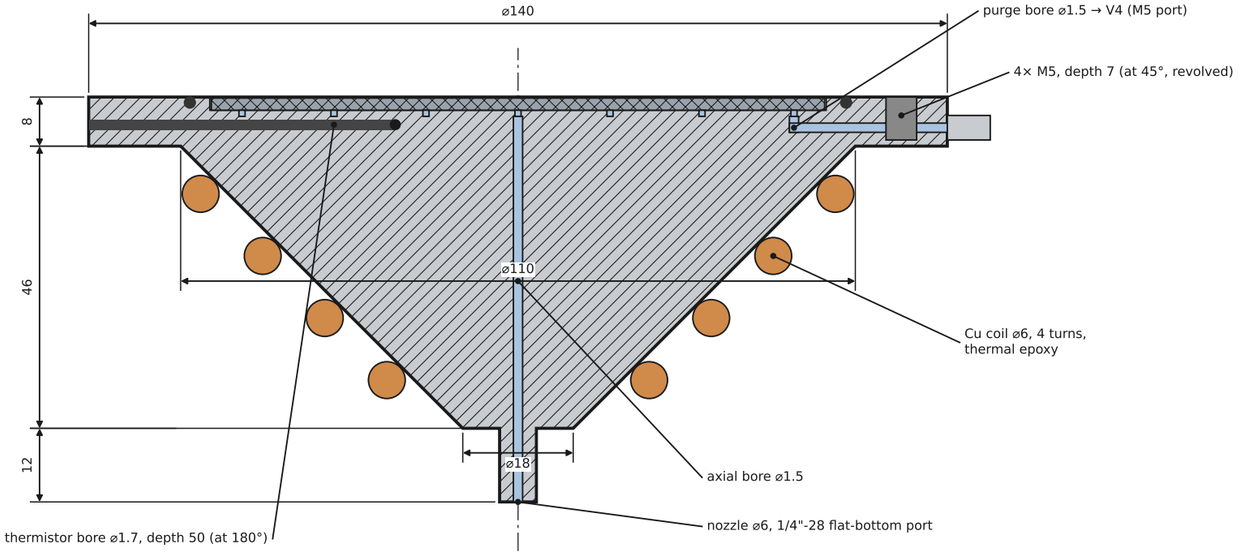
1 Arc length measured on ⌀110 from 0°; z from the tube end face (= membrane top).
2 Drill TDR holes on CNC with a fixture: rods are parallel to the x axis, not radial.
3 Seal glands with silicone (low-temperature grade); check at −10 °C before use.
4 Inner wall: thin epoxy + 0.3–0.5 mm sand coat against wall flow; mask holes.
5 Holes in the insulation jacket follow the same pattern.

PMMA tube ⌀110/100 × 125, seen from outside; angle increases to the right

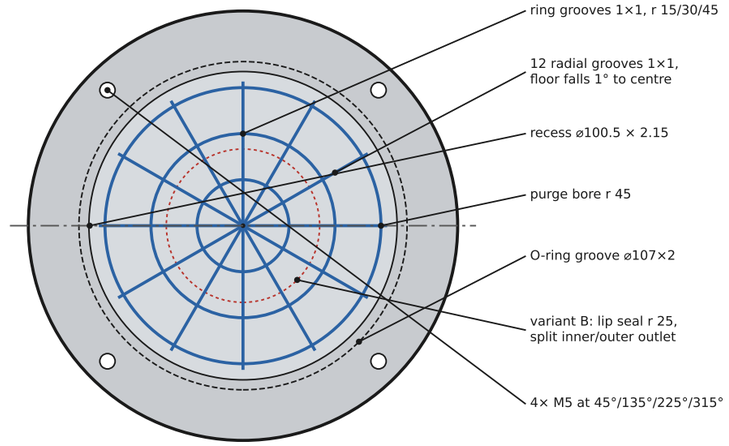
Column tube – developed outer surface

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	**1 : 1**	drawing no.	**E6-V03**	sheet 3 / 7

SECTION B–B 1 : 1

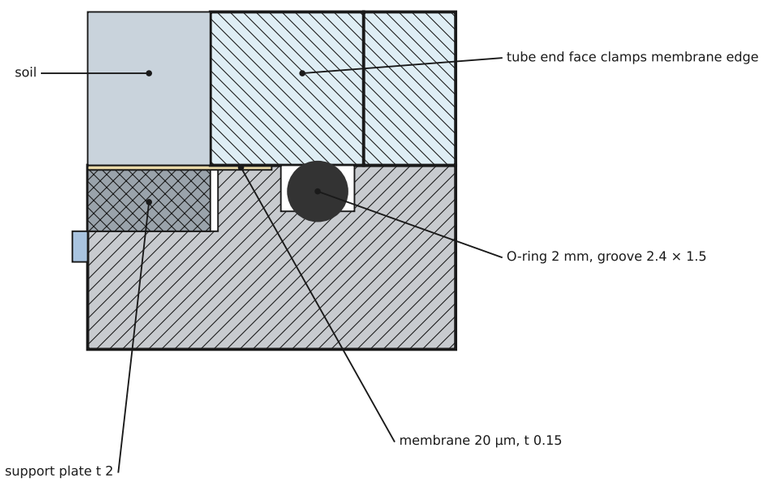


VIEW C – top face (membrane removed) 1 : 2



Why a solid cone
The external funnel keeps the drip point on the axis, but the inside is solid, so the drainage volume is grooves 1.1 mL + bore 0.1 mL (< 1 fraction) instead of the 120 cm³ air space of a hollow funnel. The membrane lets the bottom be held at a set suction h_b = 0 … −30 cm (overflow cup height, E6-V06); h_b = 0 reproduces a seepage face. The Cu coil keeps the cone at −3 °C while freezing and at +0.3 °C during outflow.

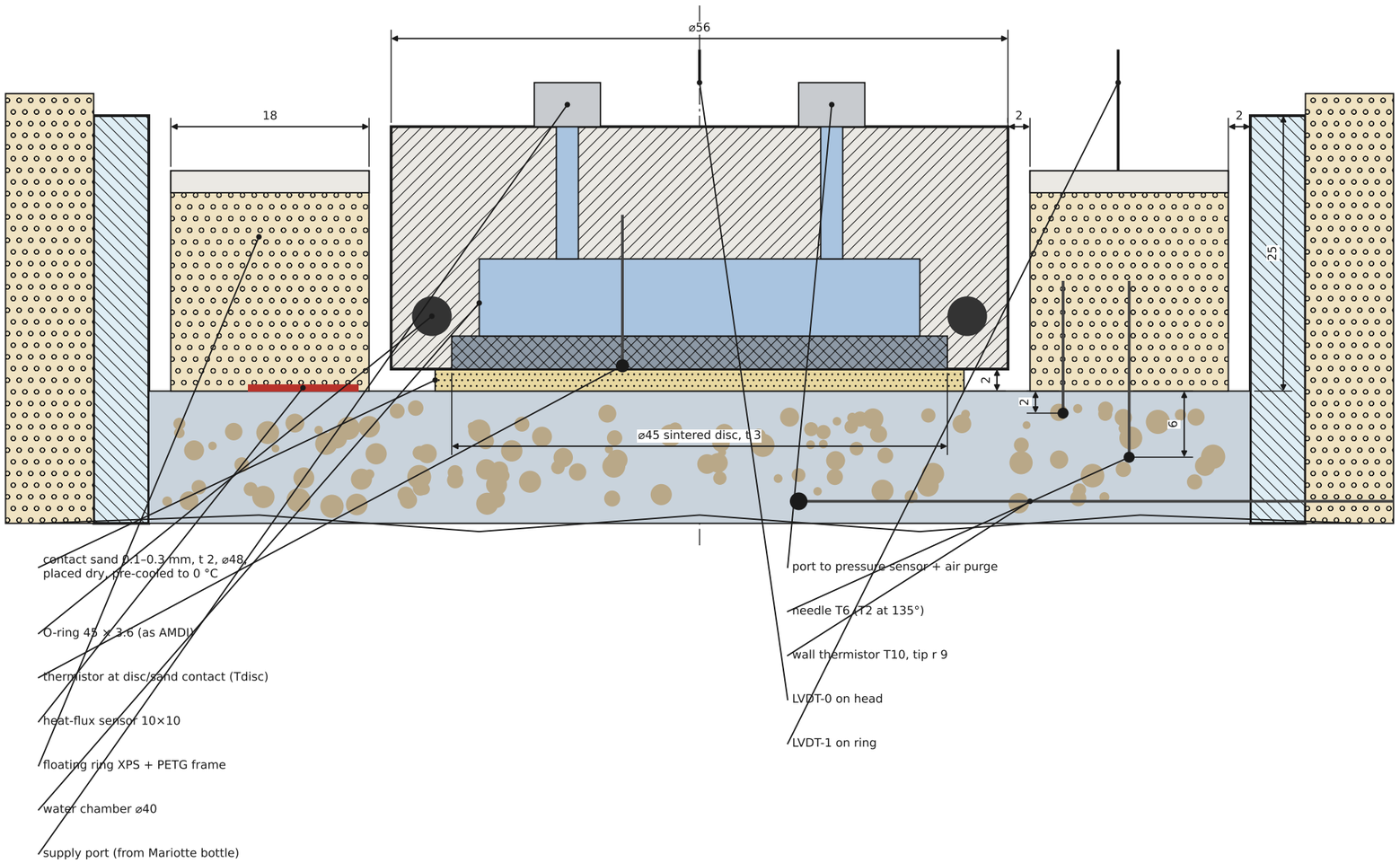
DETAIL D – membrane clamp 5 : 1



solid Al cone: external 45° funnel, internal drainage volume ≈ 1.2 mL

Cone outlet with suction membrane

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	**1 : 1 / 5 : 1**	drawing no.	**E6-V04**	sheet 4 / 7



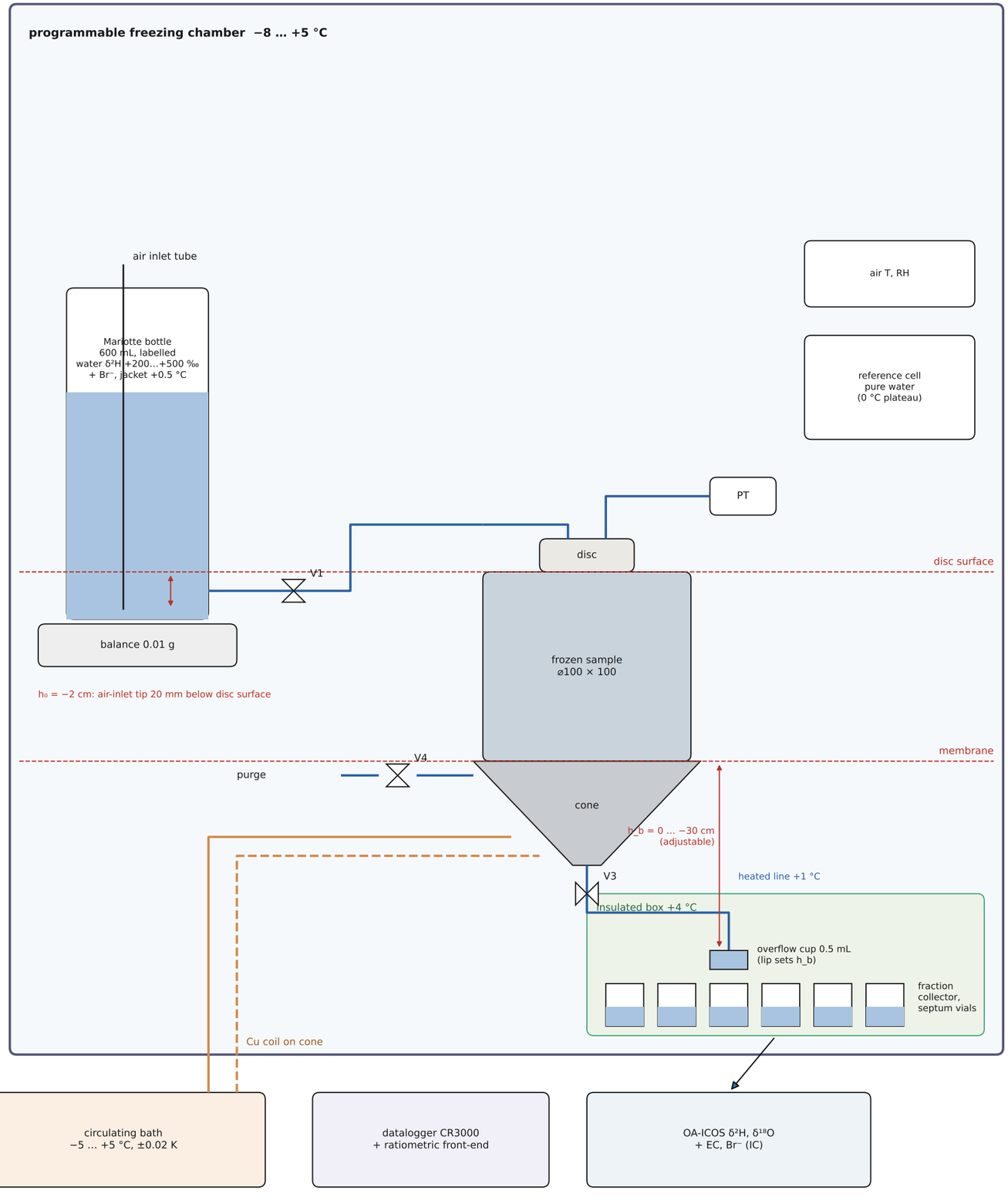
Head changes against the AMDI repository

- Keep the printed lower body, disc seat and O-ring; drop the on-board Mariotte tube.
- Two top ports: supply from the external Mariotte bottle on a balance, and a port to the differential pressure sensor that also serves as air purge.
- Head mass full ≈ 60 g (≈ 0.4 kPa on the disc) instead of ≈ 300 g (≈ 2 kPa).
- Gaps of 2 mm let head and ring float freely with frost heave / thaw settlement.
- Supply water and head stay at +0.5 °C; log Tdisc to detect freezing in the sand.

AMDI disc (Kubát et al. 2025, github.com/kubatjanf/AMDI), low-profile head

Disc–soil interface detail

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	**2 : 1**	drawing no.	**E6-V05**	sheet 5 / 7



Valves and boundary conditions

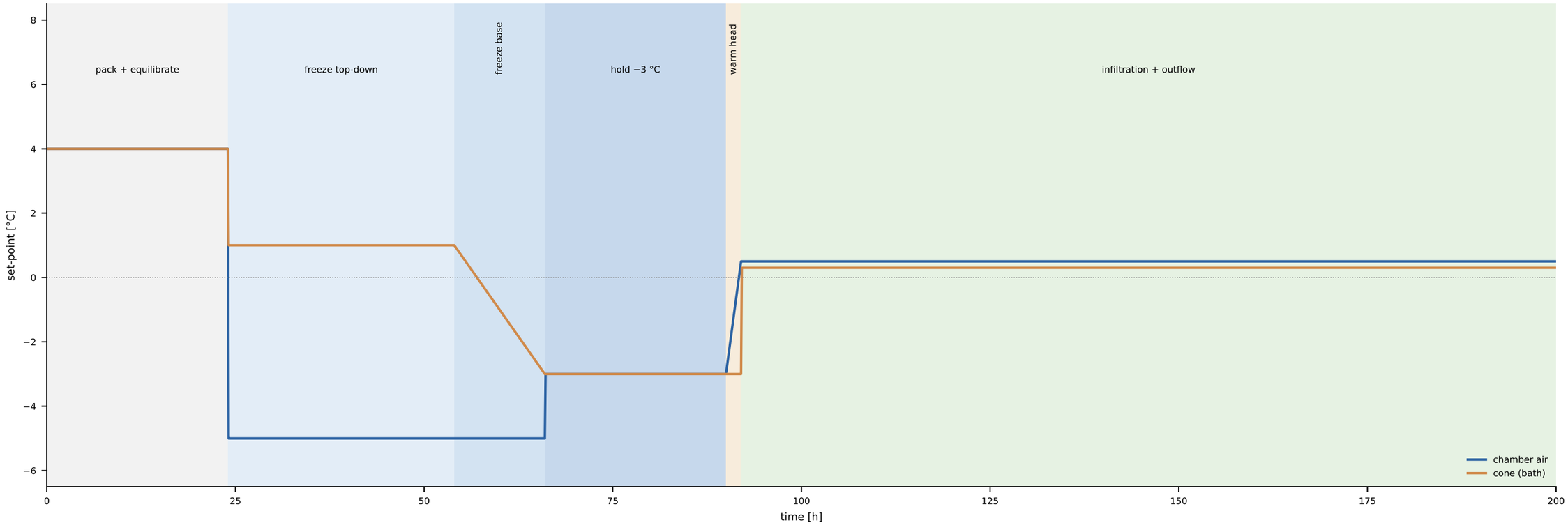
item	setting	note
V1	supply on/off	closed until infiltration start
V3	outflow	closed while freezing (stops cryosuction from the outlet line), opened at start
V4	air purge	flush grooves before freezing
top	$h_0 = -2$ cm	set by Mariotte air inlet; PT logs real h_0
bottom	h_b = 0…−30 cm	set by overflow-cup lip; h_b = 0 → seepage face (original design)
sides	≈ adiabatic	Kaiflex 30 mm; check with T10…T90
cone	T_b	−3 °C freezing, +0.3 °C during outflow

```
Balances
water:   in (balance) − out (fractions) = ΔS; ΔS checked by TDR (liquid)
         + gravimetric sectioning at the end (total = liquid + ice)
tracer:  δ²H label, Br⁻; δ¹⁸O left unlabelled → indicator of ice–water
         fractionation and of pre-event water
heat:    T profile + 2 heat-flux sensors + cone bath power
```

not to scale except the marked elevations

Hydraulic and thermal scheme

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	–	drawing no.	**E6-V06**	sheet 6 / 7



step	time	action	go / no-go
1	0–24 h	pack in 10 mm lifts, wet to θ ≈ 0.30 with background water; flush grooves (V4), V3 closed	T profile uniform ±0.05 K
2	24–54 h	air −5 °C, cone +1 °C: freezing front moves down; V3 closed	front at T90 depth
3	54–66 h	cone ramp to −3 °C at 0.3 °C/h	all T < −2.8 °C
4	66–90 h	isothermal hold −3 °C; record heave (LVDT)	drift < 0.02 K/h
5	90–92 h	air to +0.5 °C; place dry, pre-cooled sand + head; fill supply	Tdisc logged
6	92 h →	open V1; cone +0.3 °C, open V3; collect 2–5 mL fractions; run until ≥ 1.5 PV (≈ 350 mL) out	δ^2H plateau or 200 h
7	end	close V1, V3; section in 10 mm slices: gravimetric θ, δ^2H, Br^-	mass balance ±2 %

one run ≈ 7–9 days; times are indicative

Run protocol – set-points

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	–	drawing no.	**E6-V07**	sheet 7 / 7